

System	SciAD	SciAD-dedupe
Baseline	59.73	59.97
GAD*	81.90	-
SIF	88.11	89.13
XLM paraphrase	89.42	90.89
DistilRoBERTa paraphrase	89.20	90.56
DistilRoBERTa MS Marco	88.48	89.78
DistilBERT Quora	86.09	86.34
RoBERTa SciAD	88.18	89.46
XLM paraphrase → AuxAD	83.75	83.04
Ensemble	91.22	93.15

Table 3: Acronym disambiguation performance of various systems on the SciAD dataset. Results for the GAD model were taken from ? which used the test dataset, while the other scores were computed for the development dataset. Models below the line are the methods we tested, which represent a combination of pretrained sentence transformers and models we trained ourselves, with “→” denoting fine-tuning.

model, we found that it was valuable to include as a member of the ensemble. Our ensemble achieved an F1 score of 91.58 on the test dataset.

What is also interesting to see is that the systems tend to perform better on the SciAD-dedupe than SciAD, which is counter-intuitive considering the fact that leaking training data into the testing data should theoretically drive up performance scores. To investigate this, we extracted the subset of the SciAD development dataset that was duplicated from the training dataset, and measured the performance of a system that uses the most frequent long form of training examples with the same duplicated tokens. This method achieves an F1 score of 89.41, which is surprisingly lower than the F1 score of our models on the deduplicated data. This subset of the development dataset that repeats sentences from the training dataset thus seems to be quite noisy.

Figure 1 shows the distribution of cosine similarity scores between development examples and the testing example inferred to be most similar within the training dataset for our ensemble on SciAD. The distribution of scores for correct examples and incorrect examples are shown and normalized separately. We can see visually that $p(M|-C) > p(M|C)$ where M indicates that our development example found a perfect match in the training dataset and C indicates that our predicted expansion is correct. We can also see that if we ignore the perfect matches, correct predictions tend to have higher similarity scores than incorrect predictions, suggesting that our model can trade off recall for boosts in precision by using the similarity scores as a threshold.

Error Analysis We performed a small-scale error analysis by looking at a random sample of 50 mistakes made by the ensemble on the SciAD development dataset. Of those mistakes, 18 were genuine mistakes made by the model, and 32 were mistakes made by human annotators. 10 of

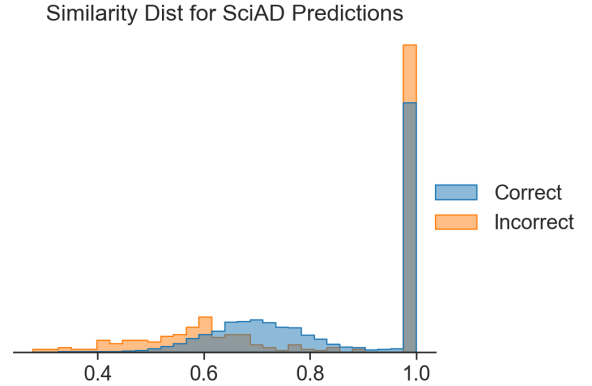


Figure 1: Similarity score distributions for predictions on SciAD by our ensemble. Each score represents the cosine similarity between an example in the development dataset and its closest example in the training dataset. The blue distribution is for examples that were judged to be correct, and the orange distribution is for examples that were judged to be incorrect. Both distributions were normalized.

the model mistakes came from semantically similar sentences across development and training having different labels, which highlights a limitation of this approach. 8 of the model mistakes were from the lack of similar training examples for a given development example, which could be potentially fixed given a larger training corpus. 30 of the human mistakes came from conflicting labels in duplicate examples, and the remaining 2 of the human errors came from mistakes during canonicalization of long forms, such as “sum capacity” and “sum capacities” both existing in the acronym dictionary.

6 Conclusion

In conclusion, we have developed new neural models for acronym identification and disambiguation. Our acronym identification model uses a transformer followed by linear projection, and our acronym disambiguation model finds similar examples with embeddings learned from Twin Networks. Both models benefited from ensembling, and both models achieve significant performance gains over the models originally proposed by ?

We introduced new datasets for acronym identification and disambiguation, AuxAI and AuxAD, which were labeled through distant supervision. We also identified a duplication issue in the SciAD dataset, and formed a deduplicated version of this dataset that we call SciAD-dedupe. We released all three of these datasets and we hope that they serve as useful tools for the NLP community.

At nam earum, ipsam nesciunt fuga ex minima error magni alias molestias, tempore placeat corrupti sunt, itaque veniam debitis nemo ut ab a?Exercitationem eaque quia velit corrupti delectus eveniet in iure quod ratione id, amet in ut tenetur necessitatibus inventore laboriosam fugit odio et quis, maiores voluptatum nihil porro, aspernatur a omnis ipsa repudiandae vel excepturi.Quos accusamus sequi, ne-

cessitatibus eos et ipsum nemo tempora nulla aspernatur
iure, doloribus delectus quidem vero numquam repudiandae
expedita culpa praesentium